

SRS v7.3 Gap Analysis – Remaining Items

 Critical / Core Feature Gaps

ID	Area	Gap	Notes
G-01	Exam Engine	Drill mode backend – POST /results/:id/drill endpoint missing. Frontend already calls it in exam-results.tsx.	type="drill" also not in DB enum
G-02	PWA	Service worker + offline caching – no manifest.json, no SW, no offline fallback	SRS §4.8 requires full PWA install + offline
G-03	Offline Sync	Dexie.js / IndexedDB offline queue – answers should survive connectivity loss	SRS §4.8.2
G-04	QR Codes	QR SVG generation endpoint – qrCodeSvg column exists but no POST /admin/questions/:id/generate-qr	Admin can't populate the field
G-05	Exam	Full-screen enforcement – SRS §3.4 requires exam to enter fullscreen and warn/auto-submit on exit	No fullscreen API calls anywhere

 High / Important Gaps

ID	Area	Gap
G-06	Study Planner	Auto-generation cron handler is empty – cron fires daily but generateDailyPlan() is a stub
G-07	Question Bank	Image/diagram support – no imageUrl on questions, no image upload in admin question form

ID	Area	Gap
G-08	PDF Notes	Inline annotation (pdf-lib) – inlineNotesTable exists for text notes but no PDF annotation layer
G-09	Performance	Rate limiting middleware – no express-rate-limit on auth or heavy routes
G-10	Observability	Sentry error monitoring – SRS §6.2 requires Sentry DSN integration

● Medium / Nice-to-have Gaps

ID	Area	Gap
G-11	External Tests	Test Tracker → Performance Trend bridge – external_tests table exists, TestTracker page exists, but dashboard weak-topics/trend doesn't factor external scores
G-12	Pomodoro	Streak sync – pomodoro sessions saved but don't update the streakDays counter on user profile
G-13	CDN	Cloudflare R2/CDN for B2 egress – B2 signed URLs serve directly; SRS §5.3 recommends CDN prox